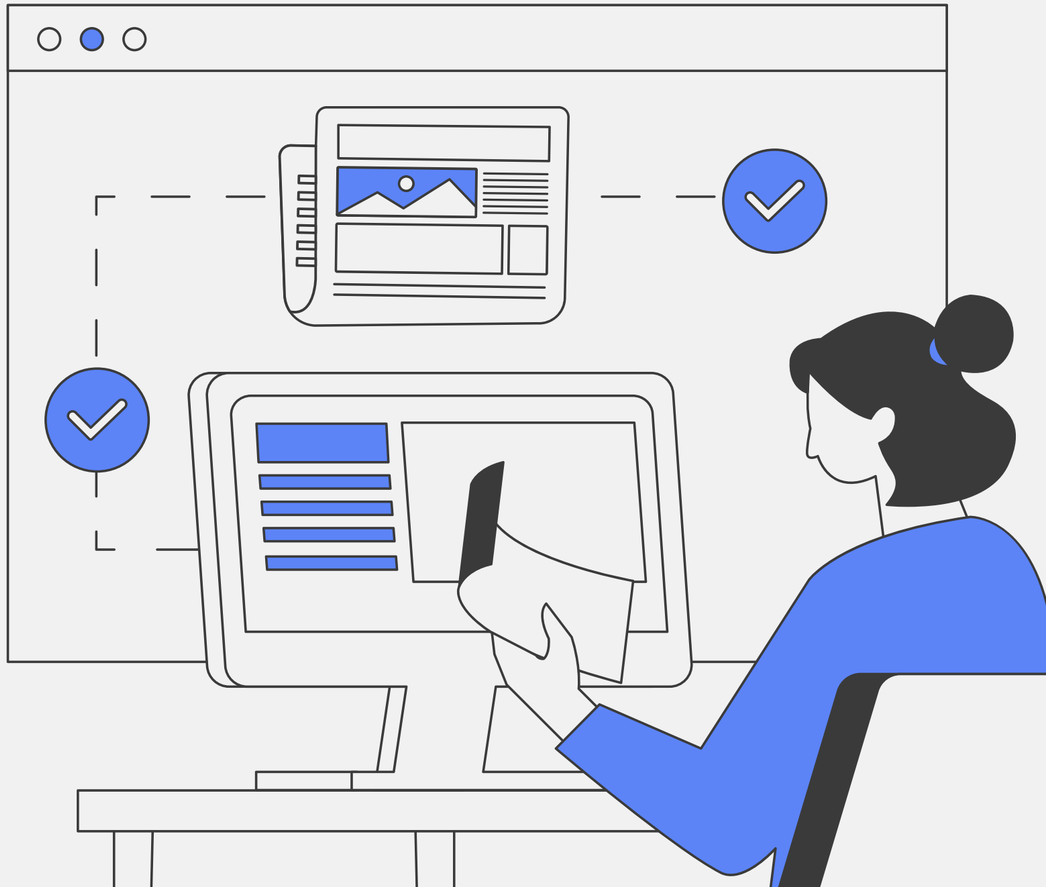
Assignment 3
Use Case 2

Group B
Umair Abdullah Muhammed
Thanh Duong Dong
Zirong Wen
Kate Durie

SOCIAL MEDIA INTELLIGENT PLATFORM

Monitoring sentiment, topics, accounts and trends using integrated Basic and Advanced NLP



WHY DO WE NEED A SOCIAL MEDIA INTELLIGENCE PLATFORM?

Problem

Social-media content is high-volume, noisy and difficult to monitor manually.

Required capabilities

- analyse public sentiment
- detect entities and social features
- identify historical trends and topics
- generate useful monitoring insight

Our Response

An integrated NLP workflow that converts raw content into structured evidence before generating an explanation.

High-volume, noisy content

↓

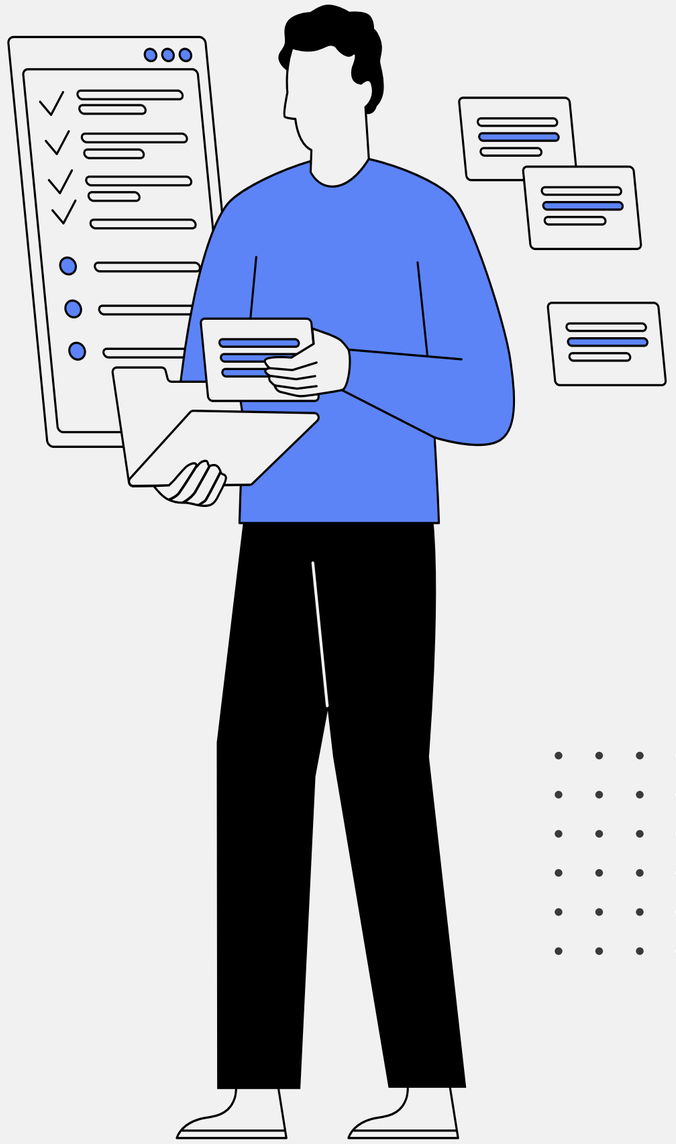
Difficult manual monitoring

↓

Integrated NLP platform

↓

Evidence-based insight



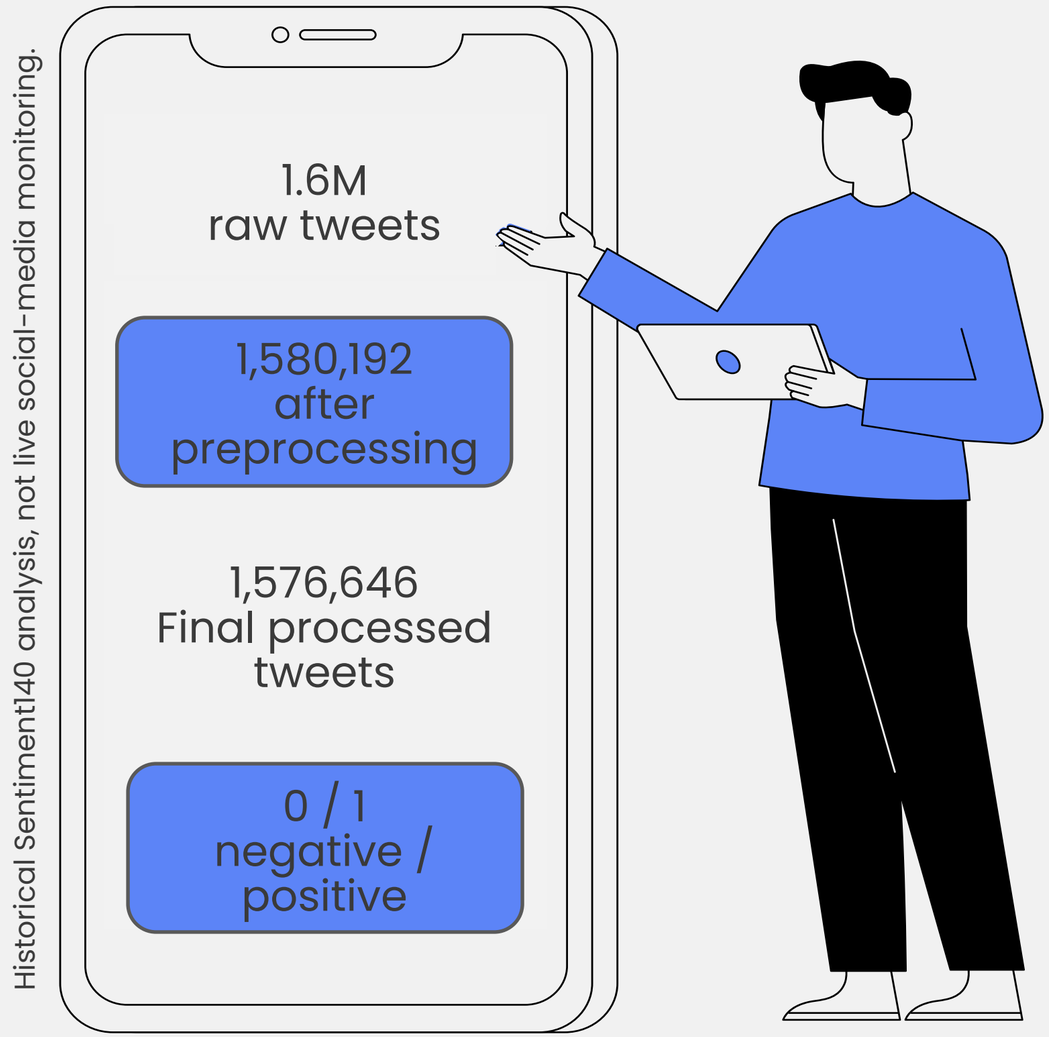
HOW DID WE BUILD A RELIABLE EVIDENCE BASE?

Why Sentiment140

- Large-scale labelled social-media dataset
- Balanced binary sentiment classes
- Original text retained for hashtags, mentions and entities
- Processed text used for sentiment modelling

Experimental Design

- 200,000-tweet modelling sample
- **80/20 train-test split**
- ~160,000 training / 40,000 held-out test
- random_state = 42
- **TF-IDF fitted on training data only**



HTML unescape
→ remove URLs and @handles
→ keep hashtag words
→ lowercase + expand contractions/negations
→ remove stopword, preserving negation
→ lemmatisation

1. Who or Where?

Named Entity Recognition (NER)

Method: spaCy

Output: people, organisations, locations

Examples detected: Dallas, Berlin, Disneyland

2. What's Tagged?

Regex extraction

Output: hashtags and mentions

- top hashtag: **#followfriday — 2,611**
- Top mention: **@mileycyrus — 4,407**

3. What's the sentiment?

Sentiment classification

Method: TF-IDF + Logistic Regression

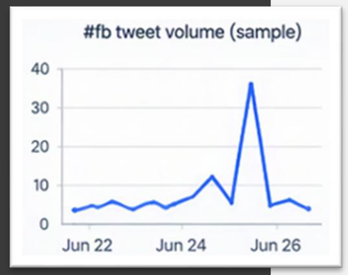
- accuracy: **0.773**
- Macro F1: **0.773**
- **40,000 tweets** held-out test tweets



4. What's changing?

Historical Trend analysis

- latest represented date: **25 June 2009**
- leading hashtag: **#fb**
- **33 tweets**



5. Who's Contributing?

Contributor activity proxy

top contributor to **#fb:**

- **Jonas_Swe**
- **13 posts**
- **0.7%** of #fb posts

6. What Theme Emerge?

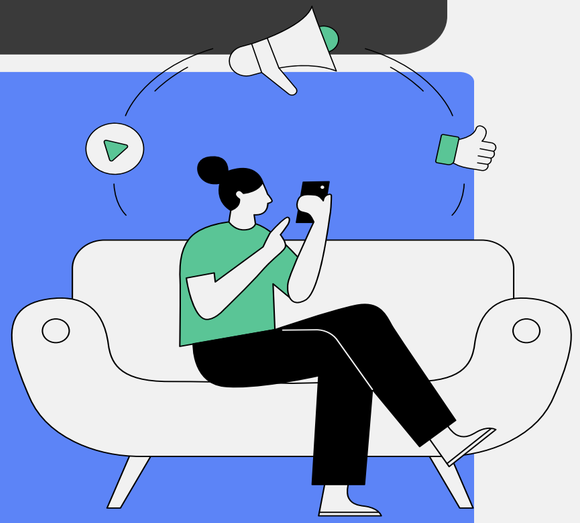
Text clustering

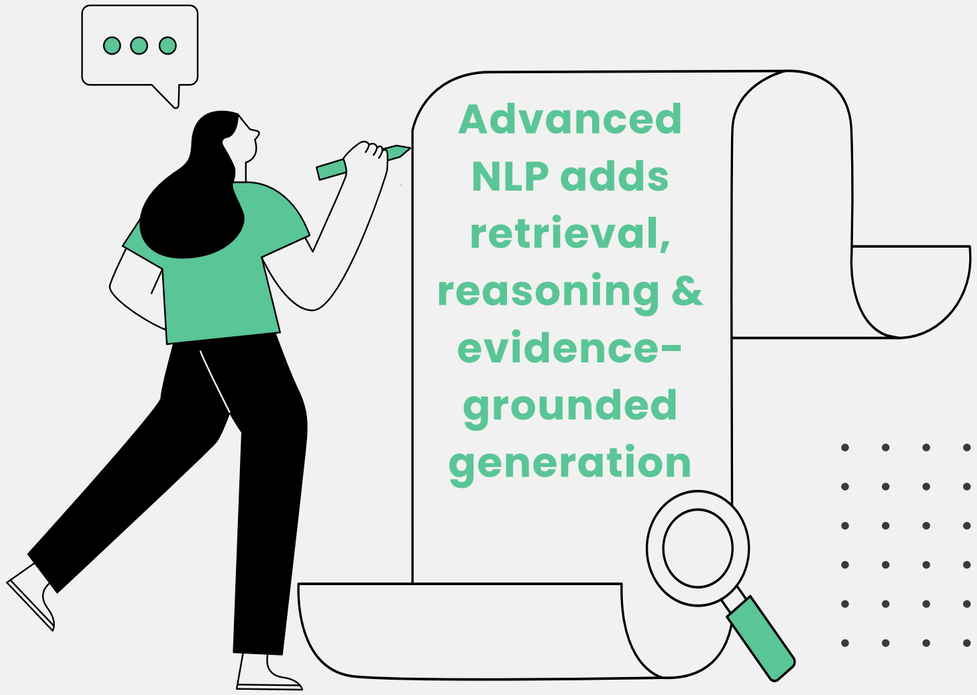
IF-IDF + MiniBatchKMeans

- 20,000-tweet sample
- **10 exploratory clusters**
- Highly imbalance cluster sizes

WHAT CAN WE MEASURE FROM A TWEET?

BASIC NLP EVIDENCE LAYER





WHAT DID ADVANCED NLP ADD?

Interpret

Foundation-model inference

- **Model**: Qwen3-0.6B
- **Mode**: Instruction-based inference
- **Role**: Interpret evidence and generate monitoring outputs
- **Used for**:
  - label candidate topics
  - interpret trend evidence
  - generate campaign recommendations

What model reasons over the evidence?

Retrieval

- **Historical source**: Sentiment140 tweets
- **Current extension**: Google News RSS → GDELT fallback

Use the #iphone experiment:

- 313 historical posts
- 100 current-news articles retrieved
- Top similarity: **0.179**
- **Retrieval worked, but relevance was weak**

What evidence do we retrieve?

Act

Trend-to-Brand generation

Evidence supplied:

- topic /hashtag, sentiment, repeated terms, historical examples, brand, audience, campaign goal

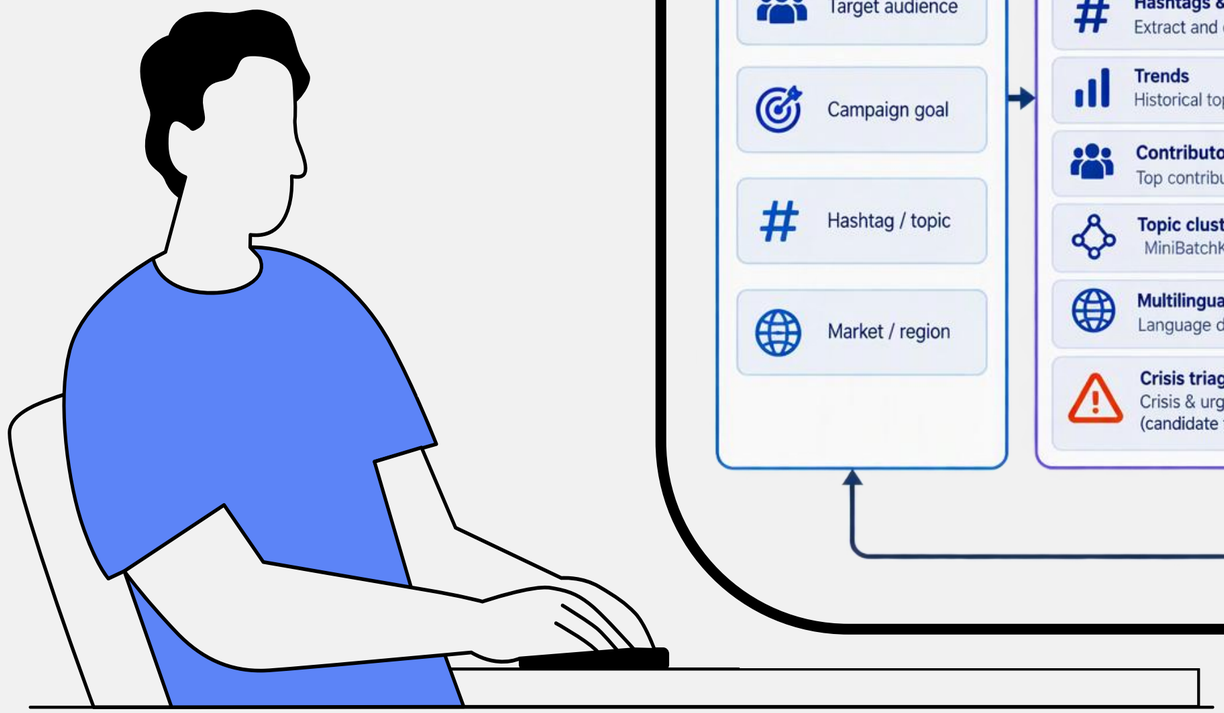
Output:

- trend explanation, brand positioning, campaign activation, sample social post, measurement ideas

Result: **Campaign brief generated**

How do we turn evidence into action?

HOW DOES THE ADV PIPELINE WORK?



Historical Social Data
Sentiment140
1.6M labelled tweets

Current News Extension
Google News RSS
GDELT (fallback)

1. INPUT
Monitoring request

- Brand / product
- Target audience
- Campaign goal
- Hashtag / topic
- Market / region

2. EVIDENCE ENGINE
Analyse and extract signals

- **Sentiment** — TF-IDF + Logistic Regression
- **Entities (NER)** — People, organisations, locations
- **Hashtags & mentions** — Extract and count
- **Trends** — Historical topic activity
- **Contributor proxy** — Top contributing accounts
- **Topic clustering** — MiniBatchKMeans (exploratory)
- **Multilingual analysis** — Language detection + sentiment
- **Crisis triage** — Crisis & urgency keywords (candidate flag)

3. RETRIEVAL & RANKING
Select supporting evidence

Combine query terms
(topic + key terms)
↓
TF-IDF retrieval
↓
Cosine similarity
↓
Top-k ranked evidence

Example result (#iphone)
313 historical posts
100 news articles retrieved
Top similarity: 0.179

Retrieval worked, but relevance was weak

4. GROUNDED GENERATION
Generate the insight

Structured evidence package
- Top posts / articles
- Sentiment and key terms
- Entities and contributors
- Trends and topics
- Language and crisis signals

↓
Grounded prompt
Instructions to use only supplied evidence
↓
Qwen3-0.6B
Instruction-based generation
↓
Monitoring / campaign insight
- Trend explanation
- Brand positioning ideas
- Campaign activation
- Sample social post
- Measurement suggestions

5. VALIDATE
Check before use

Is the evidence relevant and sufficient?

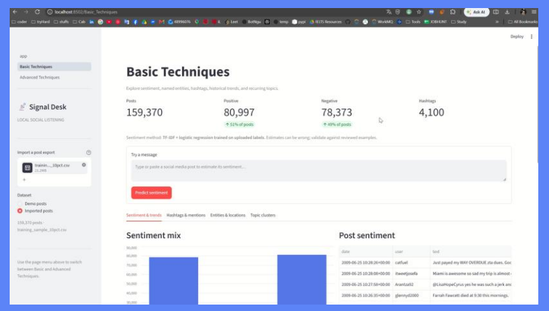
Yes → **Accept** — Use insight

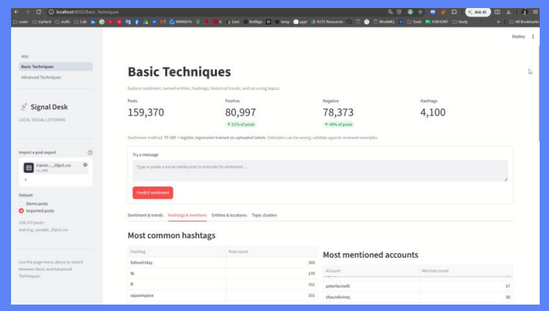
No / Uncertain → **Qualify** — Flag limitations
↓
Abstain — Evidence too weak
↓
Human review — Required for high-impact use

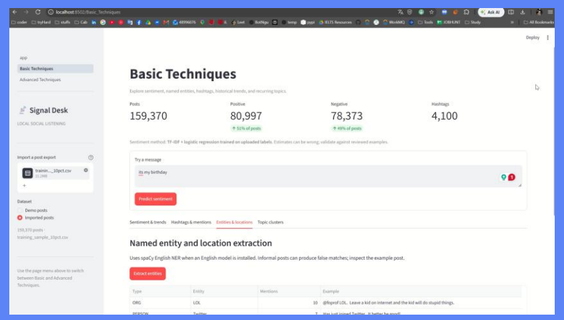
Feedback and improvement
Refine queries, thresholds and prompts based on results

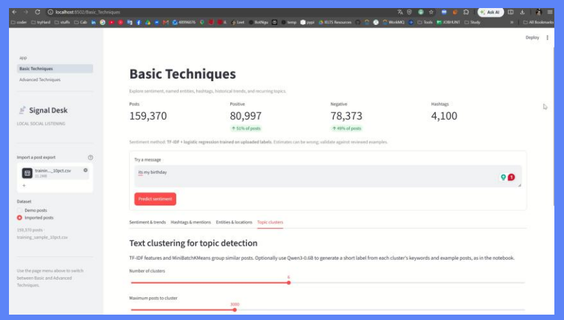
System Architecture + Workflow

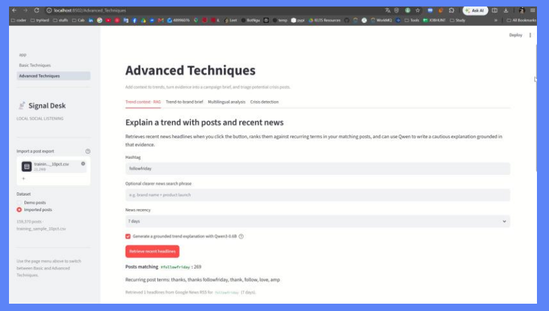
CAN WE TURN THE EVIDENCE INTO ACTION?

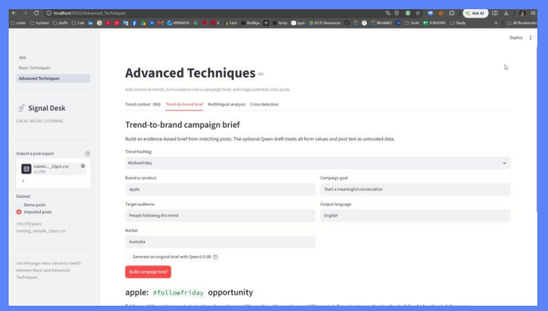


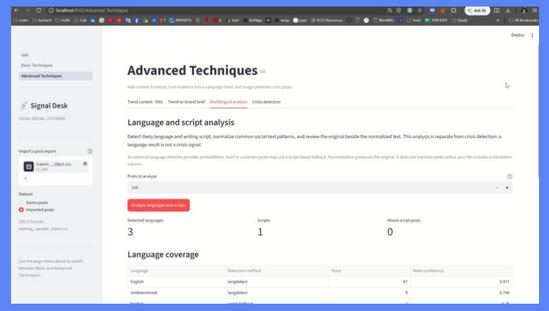


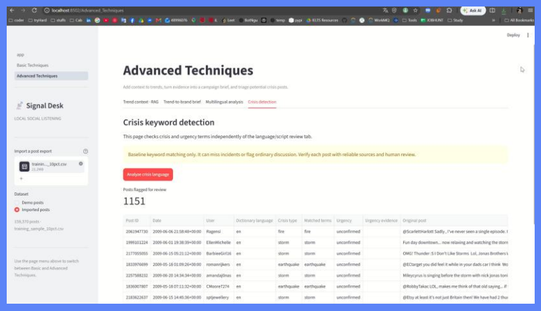












Live demo click to play ▷

DID THE SYSTEM ACTUALLY WORK?

Running successfully is not the same as performing reliably.

What worked Measurable Results

Sentiment classification
- 0.773Accuracy
- 0.773Macro-F1
- 40,000 held-out test tweets

Pipeline execution
- ✓ Social features extracted
- ✓ Historical trends calculated
- ✓ Evidence retrieved
- ✓ Campaign brief generated

What was weak Evidence Limitations

Current-news retrieval
- **0.179**
- Top displayed cosine similarity
- 100 articles retrieved, but
- semantic alignment remained weak

Text clustering
- 10 clusters produced
- Highly imbalanced distribution

Multilingual pathway
- 500 posts analysed
- 500 detected as English

Cross-language capability was not validated by this sample

What changed Methodology Decision

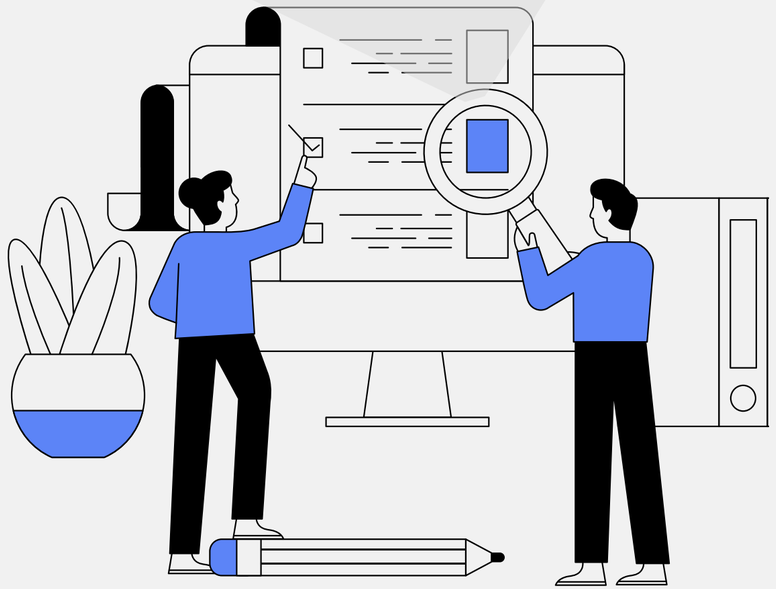
Weak retrieval
→ Add relevance threshold and abstention

Imbalanced clusters
→ Treat topics as exploratory evidence

English-only sample
→ Require a genuine multilingual evaluation set

Contributor counts only
→ Report a contributor proxy, not verified influence

Generated content exceeded evidence
→ Retain human review



WHAT DID THE ERRORS TEACH US?

01 — TOPICS
Observed: Highly imbalanced clusters
Decision: Exploratory evidence only

02 — RETRIEVAL
Observed: Weak semantic relevance
Decision: Threshold + abstention

03 — GENERATION
Observed: Output could exceed evidence
Decision: Grounding + human review

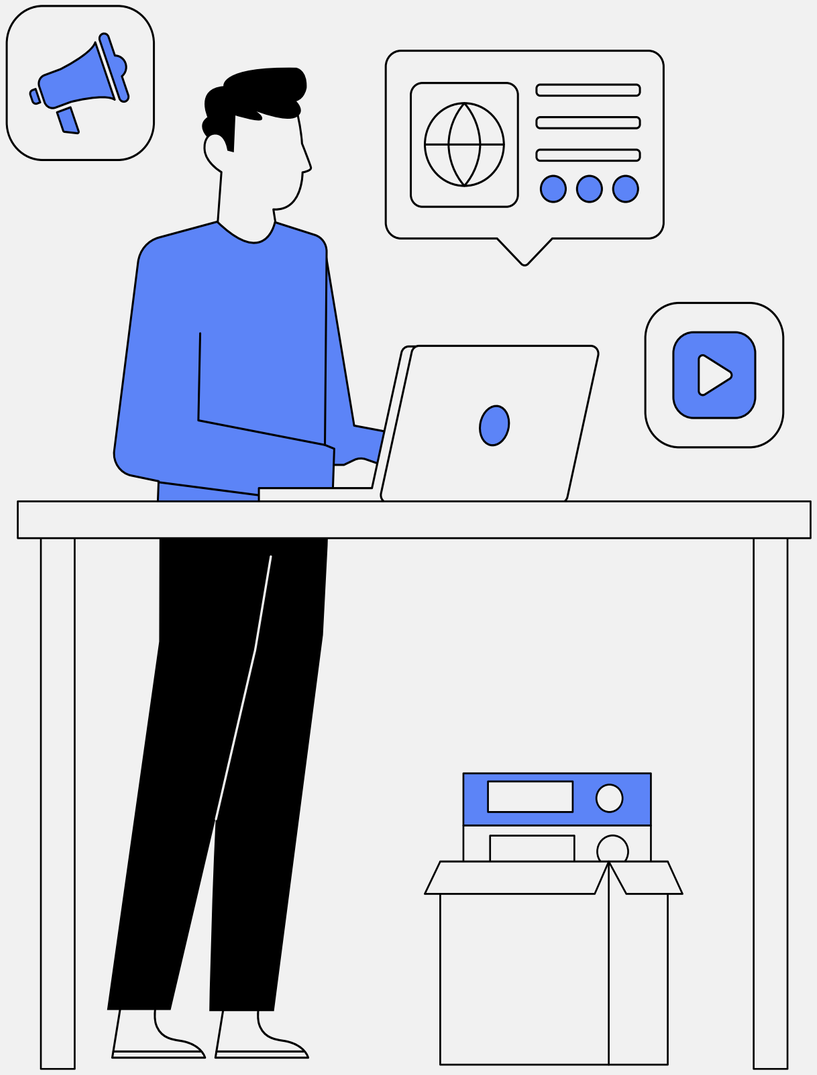
Not every technique produced the result we initially expected, and that became one of the most useful findings of the project.

Clustering generated candidate topic groups, but the highly uneven distribution meant that we could not justify treating them as reliable topic classes. Current-news retrieval successfully returned content, but weak relevance showed that retrieval alone does not guarantee useful evidence. Generated recommendations were fluent, but could still extend beyond the evidence supplied.

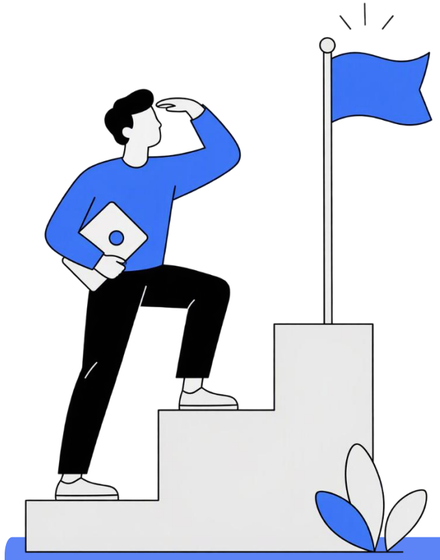
Rather than hiding these failures, we changed how the system used and described those outputs.



WHAT WOULD WE CHANGE NEXT?



Observed limitation	What we would change	Expected benefit
Historical social-media data	Add an authorised contemporary social-media source	Enables more relevant and current trend monitoring
Binary sentiment labels	Extend evaluation to neutral and mixed sentiment	Better represents real social-media discussion
Contributor activity ≠ influence	Add engagement, follower and network evidence	Supports more defensible influencer analysis
Highly imbalanced topic clusters	Test stronger text representations and validate topics with human-reviewed examples	Produces more reliable topic signals
Weak retrieval relevance	Improve query construction, add a minimum relevance threshold and allow abstention	Prevents weak evidence being treated as useful context
Multilingual analysis not fully validated	Evaluate on a genuine multilingual, human-reviewed test set	Provides defensible evidence of cross-language performance
Crisis triage not formally evaluated	Define the crisis taxonomy and test against human-labelled crisis/non-crisis examples	Allows false positives and missed crisis candidates to be measured



Three final finding

1 **Evidence before explanation**
Basic NLP created measurable and inspectable signals.

2 **Complexity did not guarantee quality**
Clustering, retrieval and generation still required evaluation.

3 **Claims must match the evidence**
Weak evidence required qualification, abstention or human review.

FROM SOCIAL CONTENT TO DEFENSIBLE INSIGHT

We built more than a collection of NLP models.

Basic NLP created measurable evidence. Advanced NLP added context and generation. Evaluation showed us when that evidence was strong enough to support a claim, and when it was not.

The result is a monitoring workflow designed around traceability, appropriate caution and human review.

QUESTIONS?

APPENDIX A Induvial contribution

Team member	Initial prototype / technical areas	Final integration contribution
Kate Durie	Data exploration and preprocessing, regex extraction, traditional sentiment models, Transformer comparison, fine-tuned DistilBERT; trend analysis, clustering, grounded LLM prototype	Results synthesis, system architecture, report integration, presentation design
Zirong Wen	Basic-model comparison, NER evaluation design, foundation-LLM sentiment experiments, zero-shot/few-shot prompting, prompt validation	Advanced NLP analysis, prompt-engineering evidence, Multilingual trend analysis
Thanh Duong Dong	NER, regex features, sentiment classification, historical trends, contributor analysis, clustering, multilingual/current-content extensions	Consolidated notebook approach, local file-upload application, system demonstration
Umair Abdullah Muhammed	TrendPulse prototype, preprocessing, sentiment modelling, trend/rising-hashtag analysis, contributor/influencer proxy, clustering	Crisis-triage investigation, evaluation, limitations and final reflection section